AI Playbook and Toolkit for Public Health Departments

AI Project Management and Implementation Planning

PGM 220

AI projects require disciplined implementation planning because they combine technology, data, governance, workflow change, training, validation, procurement, security, and evaluation. A project that appears small, such as adding an AI summarization tool, may still require data access decisions, system integration, human review design, communications planning, and post-launch monitoring. Public health agencies need project management methods that account for these dependencies before implementation begins. This module teaches program managers, product owners, implementation leads, and technical staff how to structure AI initiatives from concept through deployment and operations. The focus is on practical project controls: charters, milestones, governance gates, risk logs, stakeholder engagement, testing, training, launch readiness, and transition to operations. Strong AI project management helps agencies avoid pilots that never scale, tools that are deployed without ownership, and vendor-led projects that outpace governance or public health readiness. It also helps leadership understand what resources are needed for safe and sustainable implementation.

Level	applied/foundational
Primary track	Program Management Role-Based Track
Appears in tracks	program-management, operations-and-data-quality, technical-architecture, public-health-executive-leadership

Learning Objectives

- Define the major phases of an AI implementation project.
- Develop an AI project charter that includes scope, objectives, roles, risks, dependencies, and governance gates.
- Identify implementation dependencies across data, systems, privacy, security, procurement, workflow, training, and evaluation.
- Create a launch readiness checklist for an AI-supported workflow.
- Plan transition from pilot to operations, including ownership, monitoring, maintenance, and sustainability.

Module Overview

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Public Health Example

A public health agency plans to pilot an AI tool that drafts summaries of incoming case reports. The project plan identifies governance approval, data access, system integration, validation, user training, SOP development, incident response, and monitoring as required workstreams. The project cannot launch simply because the tool works in a demonstration. It must pass readiness checks showing that staff know how to review the output, errors can be reported, and the tool is monitored after launch.

Topic Deep Dive

AI project planning should begin with a charter that defines the use case and sets boundaries. The charter should explain the public health problem, intended users, expected benefits, data involved, systems affected, approved use, prohibited use, and success measures. It should also identify the sponsor, product owner, implementation lead, technical lead, privacy or legal contacts, security contact, evaluation lead, and operational owner.

Implementation should be organized into phases. Early phases usually include discovery, intake, feasibility review, governance approval, and business analysis. Middle phases include design, procurement or configuration, data access setup, integration, validation, workflow design, SOP development, and training. Later phases include user acceptance testing, launch readiness review, deployment, monitoring, and transition to operations. Governance gates should be placed between phases to prevent premature launch.

AI project risk management should be more detailed than a general project risk log. Risks may include poor data quality, unavailable interfaces, unclear authority, model performance problems, vendor delays, privacy concerns, security vulnerabilities, staff resistance, workflow burden, community trust concerns, and sustainability gaps. Each risk should have an owner, mitigation plan, status, and escalation threshold.

Transition to operations is often neglected. A pilot may show promise, but routine use requires support, monitoring, maintenance, budget, staff training, user feedback, incident response, and periodic review. The project plan should define when the implementation team ends and who becomes responsible for ongoing operations.

Risks, Failure Modes, and Guardrails

Risks and failure modes include the following:

Pilots launching without governance approval or operational ownership.

Project plans overlooking data, privacy, security, workflow, or training dependencies.

Vendors controlling implementation timelines without agency readiness gates.

No transition plan from pilot to routine operations.

Success measured only by deployment rather than public health value and safe use.

Recommended guardrails include the following:

Use an AI-specific project charter and risk log.

Define governance gates before implementation begins.

Create separate workstreams for data, technology, governance, workflow, training, validation, and evaluation.

Require launch readiness review before deployment.

Document transition to operations and sustainability responsibilities.

Reflection Questions

Where does this topic appear in your agency, program, or workflow?

Which staff roles would need to understand or apply this concept before AI use is approved?

What could go wrong if this topic is not addressed before implementation?

What evidence would governance, leadership, or operations staff need before moving forward?

Which policy, data, equity, privacy, security, or workflow questions remain unresolved?

Practical Exercise

Create an AI project charter and implementation plan for one proposed AI-supported public health workflow. Include project purpose, scope, objectives, roles, governance gates, milestones, workstreams, dependencies, risks, launch readiness criteria, and transition-to-operations plan.

Expected Artifact or Evidence

AI project charter

Implementation workplan

AI-specific risk log

Launch readiness checklist

Transition-to-operations plan

Expected Assignment

- AI project charter
- Implementation workplan
- AI-specific risk log
- Launch readiness checklist
- Transition-to-operations plan

Knowledge Check

- 1. What distinguishes AI project management from ordinary software implementation?
- 2. What is a governance gate?
- 3. Which item belongs in an AI project charter?
- 4. Why is transition to operations important?
- 5. What is strong completion evidence?

References and Resources

- NIST Artificial Intelligence Risk Management Framework: https://www.nist.gov/itl/ai-risk-management-framework
- NIST Generative AI Profile: https://www.nist.gov/publications/artificial-intelligence-risk-management-framework-generative-artificial-intelligence
- CDC Public Health Data Strategy and Data Modernization resources: https://www.cdc.gov/public-health-data-strategy/php/about/index.html
- OWASP guidance for LLM application security
- Agency policies for privacy, cybersecurity, records, procurement, accessibility, and public communications
- Relevant public health program SOPs, data governance documentation, and implementation plans